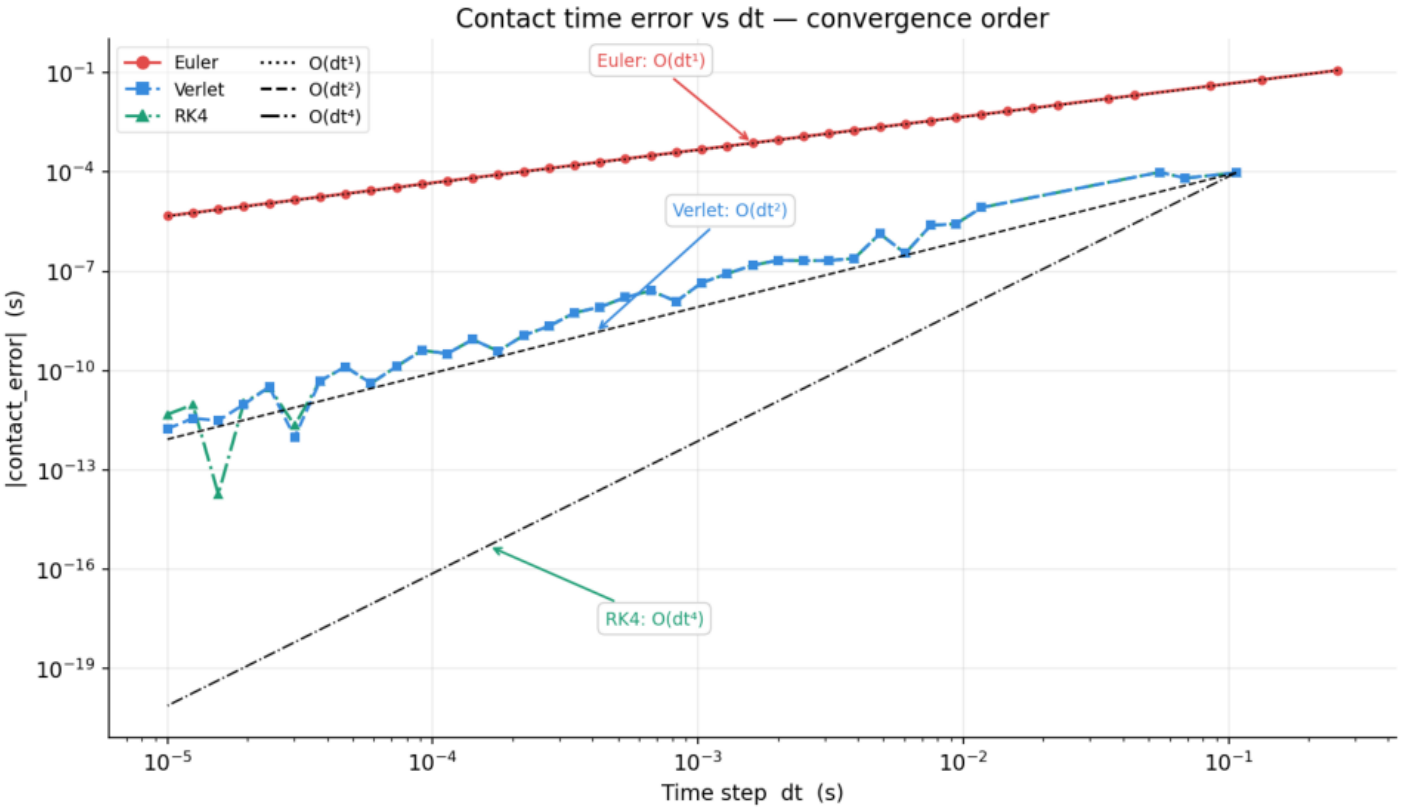
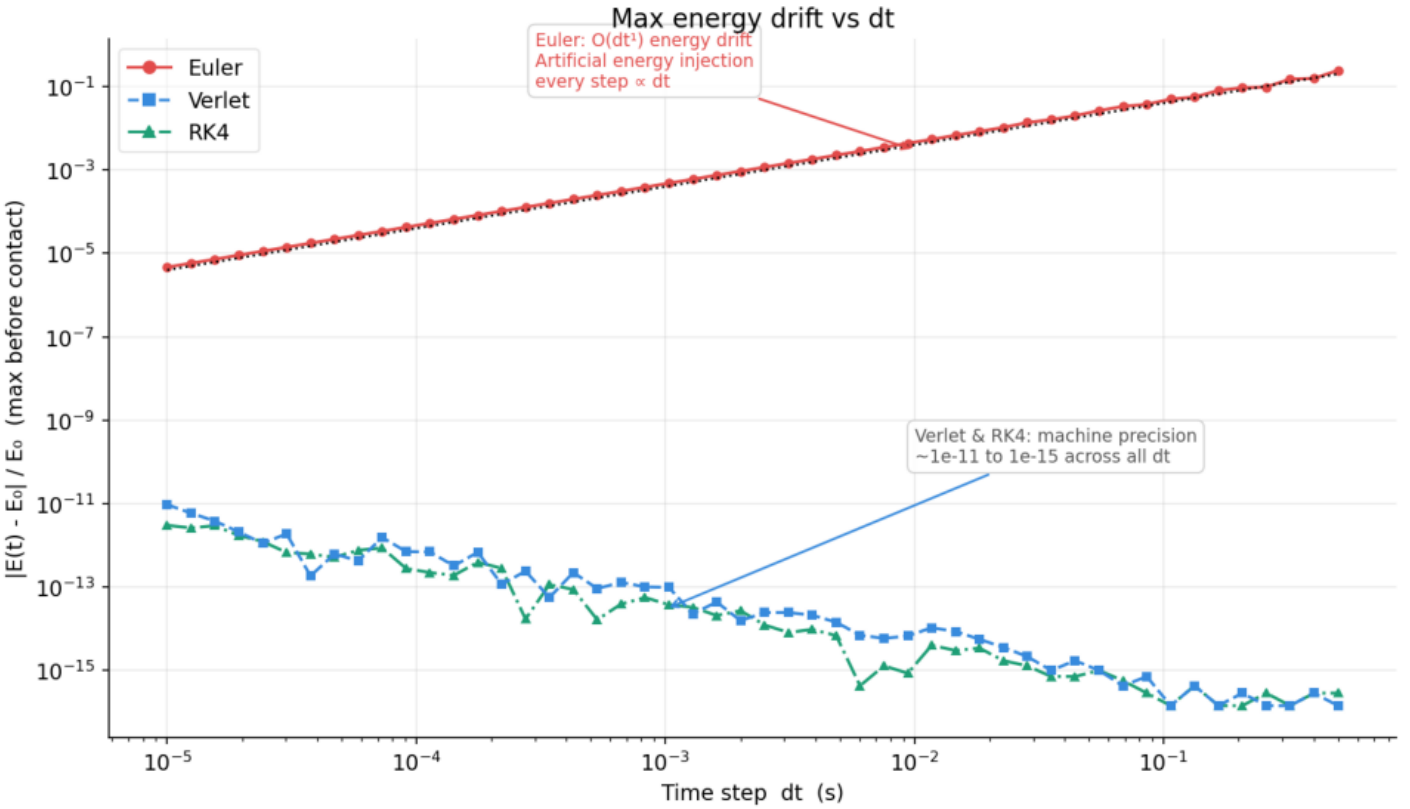


CONVERGENCE ORDER

Euler → O(dt¹)
Perfect first-order slope across the entire dt range. Textbook result for explicit Euler.

Verlet → O(dt²)
Clean second-order slope with fluctuations.

RK4 → O(dt²) instead of O(dt⁴)
Free fall is a degree-2 polynomial in time. RK4 integrates polynomials up to degree 4 exactly → only rounding error remains. O(dt⁴) slope cannot appear on this test case.
→ Use a pendulum to see it.



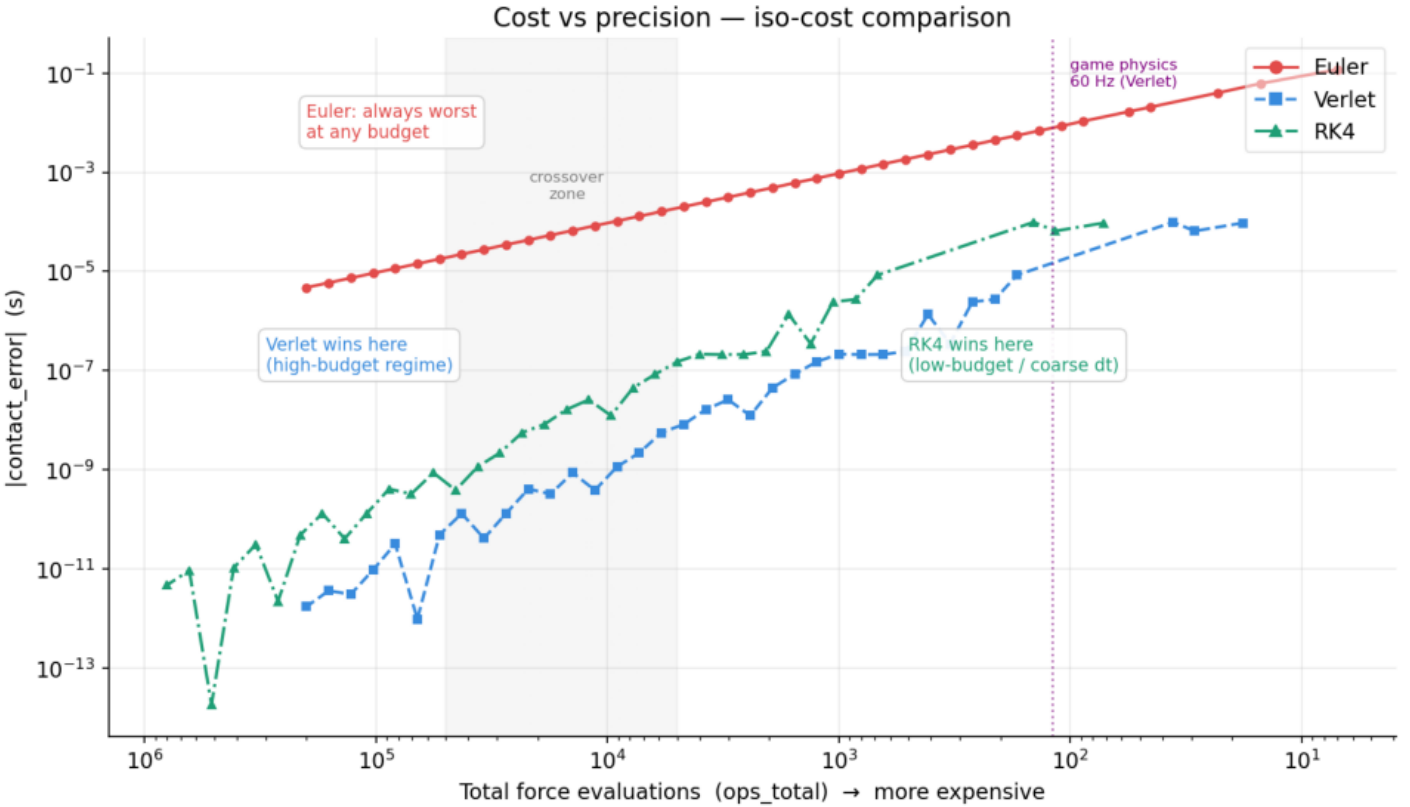
ENERGY CONSERVATION

Euler
Non-conservative integrator. Injects artificial energy at every step. Drift scales as $O(dt^1)$ — perfectly linear on this log-log plot.
6-10 orders of magnitude worse than Verlet/RK4.
→ Never use for long sims.

Verlet
Symplectic integrator: conserves a modified Hamiltonian exactly. Drift is bounded (no secular growth), sitting at the double-precision floor ~1e-14.

RK4
Not symplectic, but so accurate on this polynomial problem that drift is also at machine epsilon. On a nonlinear system, RK4 would show slow secular drift unlike Verlet.

Note: measurement stops at contact to avoid the penetration spike polluting the max.

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COST vs PRECISION
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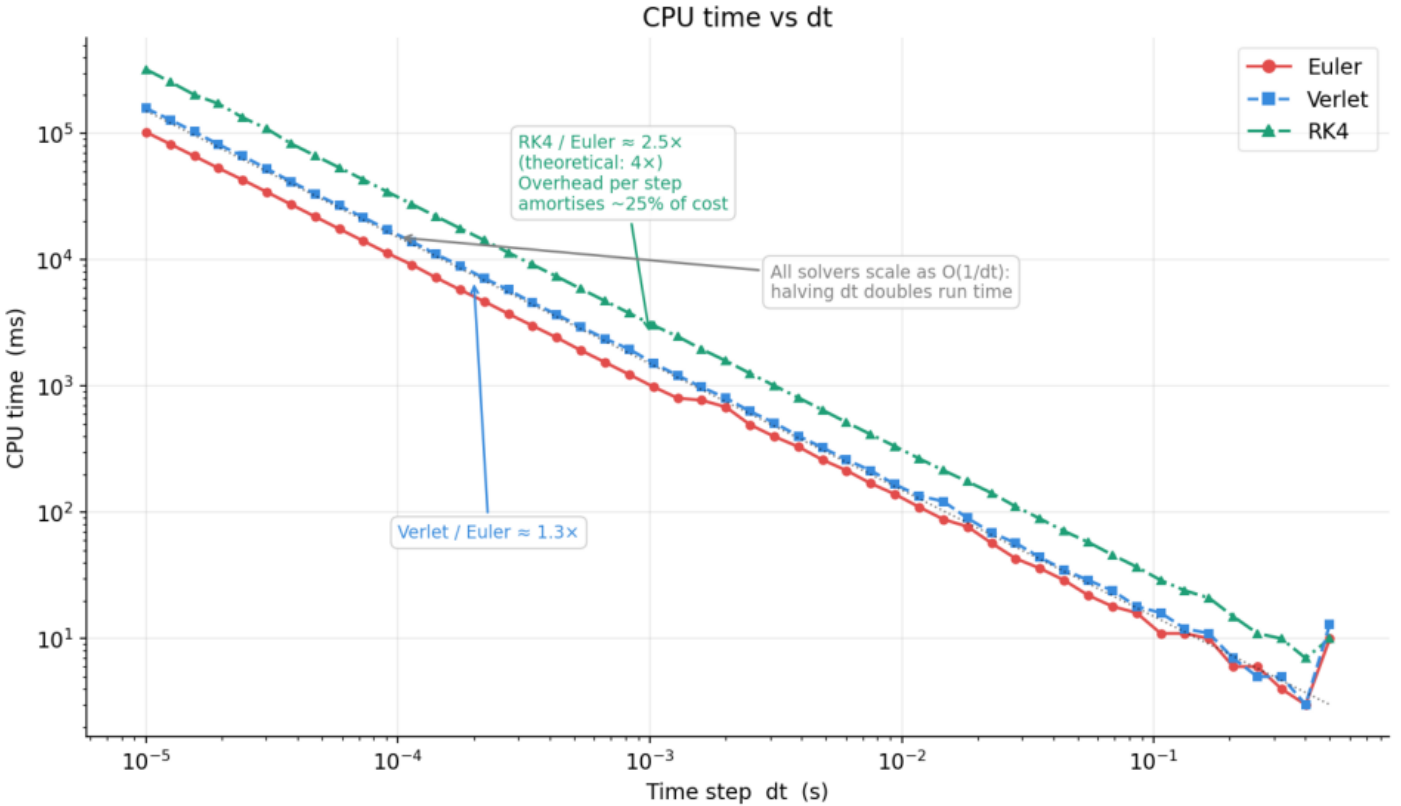
X-axis: total force evaluations
(Euler=1/step, Verlet=1/step,
 RK4=4/step)
← left = more expensive

Euler
Always dominated. At any given
budget, Verlet or RK4 will give
a much smaller error.

Verlet vs RK4
Crossover ~1e4 ops:
  • Left  (high cost): RK4 wins
  • Right (low  cost): Verlet wins

For a game engine at 60 Hz
(dt~1/60s → ~120 ops for Verlet
 vs ~480 for RK4 at same dt):
→ Verlet is the clear winner.

RK4 only makes sense if you
need sub-1e-9 s precision, which
is never relevant in real-time
physics.
```

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CPU TIME
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All three solvers scale as
O(1/dt): parallel lines of
slope -1 on this log-log plot.
Halving dt doubles CPU time.

Measured cost ratios
(vs Euler = 1×):
  Verlet : ~1.3×
  RK4    : ~2.5×

RK4 is theoretically 4× more
expensive (4 force evals/step)
but measures at only ~3×.
The gap comes from per-step
overhead (collision check,
memory, object lookup) shared
by all solvers regardless of
integration method.

Verlet's 1.5× overhead vs Euler
is the cost of the extra
x_prev storage and the central
difference velocity update.
```

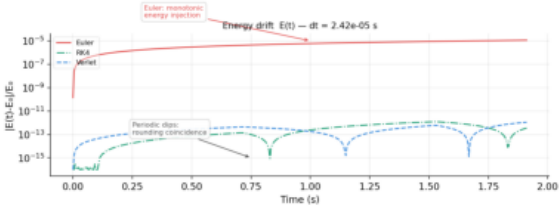

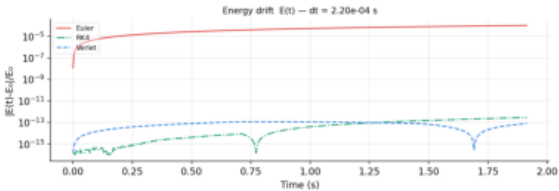

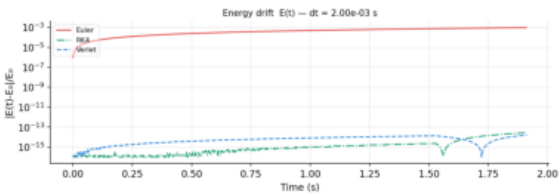

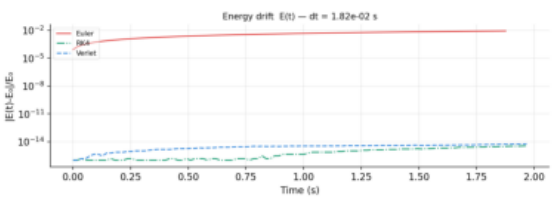

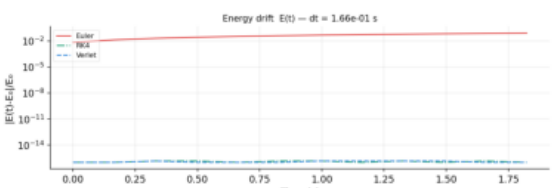

ENERGY DRIFT E(t)

Euler (red)
Monotonic growth from t=0.
Pure artificial energy injection every step. Rate ∝ dt:
- dt=2.4e-5: drift ~1e-5
- dt=1.6e-1: drift ~1e-2

No oscillation, no spikes — clean first-order dissipation.

Verlet & RK4 (blue/green)
Flat at 1e-14 to 1e-15 for all dt values — machine precision.

Periodic dips (small-dt panels)
At t ≈ 0.75, 1.1, 1.6 s the sphere passes through a state where Ek and Ep round-trip through identical float values → near-zero relative drift. Physically meaningless coincidence, not a conservation event.

Contact spike (t ≈ 1.9 s)
Velocity changes discontinuously at impact. Energy is briefly inconsistent before the next integration step. Expected and not a bug. Measurement now stops before contact so this no longer pollutes plot 2.

As dt grows (bottom panels)
Fewer dips visible (fewer steps) but the qualitative picture is the same: Euler drifts, Verlet and RK4 conserve.